

workers handle new responsibilities caused by AI chatbots.

To further understand how AI chatbots affect the nature of work, we asked respondents to describe the new job tasks they perform as a result of AI chatbots. In Appendix B.3, we categorize the free-text responses into common new tasks associated with AI chatbots in each occupation. AI chatbots have generated new job tasks for workers across all 11 occupations, with 53% to 91% of new tasks directly linked to AI use.

Figure 4 breaks down the composition of AI-related job tasks by occupation and task category. Appendix B.3 provides example tasks for each pair. About 42% of new AI-related tasks relate to generating new content, including *AI Ideation*, *AI Content Drafting*, and *AI Data Insights*. For example, marketing professionals report new tasks related to “prompting and iterating with AI to produce marketing copy, social media posts, and product descriptions.” Notably, 35% of new AI-related work is dedicated to *AI Quality Review* and *AI Ethics & Compliance*, suggesting that AI adoption shifts workers not only toward content generation but also toward supervisory roles overseeing AI outputs.¹⁰ For example, a large share of teachers report new work related to “detecting AI-generated homework.” Finally, the single most common new AI task category is *AI Integration*—that is, the integration of AI chatbots into the workplace, which accounts for about 26% of all new tasks, with the highest share observed in IT support and software development. For example, software developers report “fine-tuning AI coding assistants by providing feedback and project-specific examples,” and legal professionals describe “developing organizational AI usage policies and guidelines.”

AI chatbot workloads extend beyond direct users: Figure 3, Panel (b) shows that 4% of non-users report new workloads resulting from AI chatbots, with effects most pronounced in workplaces that have adopted chatbot initiatives. In particular, Figure B.1, Panel (b) shows that 10% of teachers who have not used AI chatbots report new AI-related

¹⁰Appendix Figure E.3 shows that *AI Ethics & Compliance* tasks are especially common among workers who have received employer-provided AI chatbot training, helping to contextualize the finding in Section 2.2 that training—when implemented in isolation—is associated with lower productivity gains from AI chatbots. In contrast, workplaces that only encourage chatbot use exhibit a higher share of “generative tasks” focused on *AI Ideation* and *AI Content Drafting*.

workloads, largely consisting of monitoring and responding to students’ AI use.

Discussion. The widespread creation of new tasks and spillovers to non-users highlight the broader workplace transformations caused by AI chatbots. These patterns support task-based frameworks of technological change, in which automation may reinstate labor demand (Acemoglu and Restrepo, 2018*a*; Autor et al., 2024), including models of “job transformation” where technologies reshape the content of jobs well before they displace them (Autor and Thompson, 2025; Freund and Mann, 2025). Workers shift not only toward content generation but also toward supervisory roles overseeing AI outputs—a pattern consistent with AI chatbots altering the hierarchical structure of production, with humans increasingly serving as monitors and validators of AI-generated work (Ide and Talamas, 2025). Finally, the high prevalence of integration tasks likely reflects an early phase of major technological change, with workers and firms still in the process of embedding AI into their workflows. These patterns could suggest that some organizations remain in the trough of a productivity J-curve (Brynjolfsson, Rock and Syverson, 2021; McElheran et al., 2025). On the other hand, as shown in Section 2.2, a substantial share of workers already report large time savings and quality improvements from using AI chatbots—especially when employers encourage and invest in the tools. In the next section, we assess whether and how these workplace transformations surface in administrative labor market outcomes.

3 The Surface: Earnings and Employment

Our preceding analysis shows that AI chatbots have generated strong currents within workplaces: employers are adopting chatbot initiatives, workers report substantial work-related benefits, and new AI-related tasks are emerging broadly. Do these currents surface in “hard” labor market outcomes such as earnings or hours? To answer this question, we link our survey responses to administrative labor market records.

3.1 Empirical Strategy

Evaluating the labor market impacts of AI chatbots raises several conceptual questions: What are the relevant treatments to consider? If the treatments have an effect, in which recorded labor market outcomes would they manifest? And how can we identify these impacts in the data?

Motivated by the dual nature of adoption—with variation across workplaces depending on their chatbot initiatives and within workplaces depending on individual use—we analyze impacts at both the worker and workplace levels.

Appendix A.2 presents a theoretical framework for how adoption translates into labor market outcomes at the worker and workplace levels, guiding the following analyses:

1. *Worker earnings:* If chatbots raise individual productivity, a natural question is whether and how these improvements translate into earnings for users relative to comparable non-users (Krueger, 1993). Wage effects depend on whether adoption is worker- or firm-driven, whether chatbot use displaces users from higher-value tasks or is valued as a nonpecuniary amenity, and the structure of wage setting (Autor and Thompson, 2025; Becker, 1964; Card et al., 2018; Rosen, 1986). Section 3.2 investigates these questions by estimating the impacts of chatbot use on workers' earnings by their intensity of use, employer initiatives, reported productivity gains, new AI-related workloads, and wage-setting systems.
2. *Workplace employment:* If chatbots reduce the need for labor, adopting workplaces might slow hiring or lay off workers. Conversely, if the tools boost productivity and spur demand, adopting workplaces could expand employment. In particular, concerns have been raised that generative AI could displace entry-level positions (Amodei, 2025; Brynjolfsson, Chandar and Chen, 2025). Section 3.4 analyzes these possibilities by estimating the effects of employer chatbot initiatives on workplace employment, early-career jobs, job churn, and incumbent worker outcomes.
3. *Job mobility:* If chatbots are especially useful for newcomers to an occupation,

adopters may be more likely to switch occupations (Autor, 2024). Section 3.3 tests this prediction, exploring heterogeneity across occupations and firm- and worker-level drivers of adoption.

3.1.1 Identification Strategy and Challenges

We use a series of difference-in-differences analyses to identify the relevant effects, comparing adopters and non-adopters before and after the introduction of AI chatbots and examining outcomes at both the worker and workplace levels. At its core, our analysis asks a simple question: have adopters fared differently from non-adopters since the arrival of chatbots?

The key identifying assumption is that, absent AI chatbots, adopters and non-adopters would have experienced similar changes in labor market outcomes. We apply this assumption at both the worker level (e.g., earnings of users versus non-users) and the workplace level (e.g., labor demand at encouraged versus non-encouraged workplaces). Appendix A.2 draws on our theoretical framework to clarify these identifying assumptions and to provide a structural interpretation of the difference-in-differences estimands.

A first concern in our analysis is that adoption may remain a relatively weak treatment: workers may struggle to use the tools effectively, and firms may provide limited guidance or infrastructure to support their implementation. Section 2.2 showed that employer encouragement is critical for unlocking both higher use and greater reported benefits. We use these “AI front-runner” workplaces—those with encouraged use—as our primary workplace treatment, since they offer a window into the effects of AI chatbots as employers embrace the tools. We also examine the effects of additional investments in employer-provided chatbots and training. At the individual level, we also analyze heterogeneity by intensity of use and reported gains.

A second concern is that adopters may have fared differently in the labor market even without AI chatbots. We take several steps to address this issue. First, we control for workers’ pre-determined characteristics—including gender, age, and labor market